

Network Reconnaissance and Security Assessment

Executive Summary

A security assessment was conducted on a virtual laboratory network to evaluate its security posture using industry-standard reconnaissance tools. Kali Linux was configured as the scanning machine, while Ubuntu Server acted as the target system. The assessment included host discovery, network mapping, port and service enumeration, operating system fingerprinting, and packet analysis. The objective was to identify exposed services, evaluate potential risks, and recommend appropriate security measures.

1. Assessment Scope

The following activities were performed:

- Host Discovery
- Network Mapping
- Port Enumeration
- Service Detection
- Operating System Fingerprinting
- Packet Analysis
- Risk Assessment

2. Tools Used

Tool	Purpose
Kali Linux	Security assessment platform
Netdiscover	Host discovery
Nmap	Port scanning, service detection, OS fingerprinting
Wireshark	Packet capture and protocol analysis
Oracle VirtualBox	Virtual lab environment

3. Scan Results

Host Discovery

The host discovery phase successfully identified all active devices within the Host-Only virtual network.

Host	IP Address	Status
VirtualBox Host Adapter	192.168.56.1	Active
VirtualBox DHCP Server	192.168.56.100	Active
Kali Linux	192.168.56.101	Active
Ubuntu Server	192.168.56.104	Active

Port Enumeration

The Ubuntu Server exposed only one TCP service.

Port	Service	State
22	SSH	Open

All other commonly scanned TCP ports were closed.

Service Detection

Service	Version
OpenSSH	10.2p1
Protocol	SSH Version 2.0

Operating System Detection

Parameter	Result
OS Family	Linux
Target OS	Ubuntu Server
Network Distance	1 Hop

4. Packet Analysis

The following network protocols were successfully captured and analyzed using Wireshark.

Protocol	Observation
ARP	Successful address resolution
ICMP	Successful ping communication
TCP	Port scan traffic observed

Protocol	Observation
SSH	Encrypted communication verified

5. Security Findings

Finding	Severity
SSH service exposed on Port 22	Medium
Limited attack surface	Low
SSH encryption enabled	Low
No unnecessary open services detected	Low
No suspicious traffic observed	Informational

6. Risk Analysis

The assessment indicates a **low overall security risk** for the evaluated virtual laboratory environment. Only the SSH service was exposed, which is expected for secure remote administration. No critical vulnerabilities or unnecessary network services were identified during the assessment. The system demonstrated a minimal attack surface and secure encrypted communication through SSH.

7. Recommendations

- Use strong passwords or SSH key-based authentication.
- Keep OpenSSH updated with the latest security patches.
- Disable direct root login via SSH.
- Perform regular vulnerability and port scans.
- Monitor network traffic for suspicious activity.
- Enable logging and periodic security reviews.

8. Conclusion

The security assessment successfully achieved the defined objectives of host discovery, network mapping, port and service enumeration, operating system fingerprinting, and packet analysis. The Ubuntu Server maintained a secure configuration with only the required SSH service exposed. Based on the observed results, the overall security posture of the virtual lab network is satisfactory, with no critical security issues identified within the project scope.